# Recording the AWS demo

A real spot instance in Sydney trains CIFAR-10. AWS sends a real interruption. SpotSentry saves in under a second, and a new instance carries on from the same step.

About 45 min in total, 30 min of it recording. Cost: a few cents.

## Screen layout

The same facts in three places at once. That is what makes it convincing.

| | |
|---|---|
| **Left** | AWS console: instance ID, **spot**, Sydney |
| **Right** | Dashboard: epoch/step, checkpoints, notice, save, restore |
| **Bottom** | Terminal: live events from S3, with instance IDs and times |

Instance IDs match across all three. Times match the taskbar clock.

## Keep off screen

- The **.aws** folder and the access key file
- Never run **aws configure**
- The console's top-right account menu (shows your email)

# The day before

## 1 Recorder ready

**OBS Studio** (has pause and microphone):

```
winget install OBSProject.OBSStudio
```

Sources > + > Display Capture. Output: mp4. Video: 1920x1080, 30 fps.

Or Snipping Tool: **Win+Shift+R**, select the whole screen (no pause).

## 2 Do Not Disturb on

So no notifications appear on the video.

## 3 AWS console tabs

Region **Asia Pacific (Sydney)**, top right.

- **EC2 > Instances**. Filter **tag:spotsentry = true**. Gear: turn on **Instance lifecycle** and **Availability zone**.
- **EC2 > Spot Requests**
- **S3** > bucket > jobs/ > cifar-aws/ > checkpoints/

# Before you press record

## T Every terminal, all three

Do this in **each** terminal:

```
cd D:\CCel
```

```
.venv\Scripts\Activate.ps1
```

The prompt must start with:

```
(.venv) PS D:\CCel>
```

Check, it must print **ready**:

```
python -c "import boto3, spotsentry;
print('ready')"
```

1 = dashboard, 2 = event stream, 3 = interrupt and FIS check.

## 0 Nothing running

In terminal 2:

```
python scripts/demo_aws.py --no-follow
```

Must say: **No running instance tagged spotsentry=true.**

# Recording

## 1 Dashboard

terminal 1

```
spotsentry dashboard --job jobs/cifar_aws.yaml
```

Put the browser tab on the right half.

## 2 Start recording

OBS: **Start Recording**.

## 3 Fresh run

terminal 2

```
python scripts/demo_aws.py --start
```

> **Say:** *"This clears the previous run and asks the Auto Scaling group for one spot instance."*

Leave it running. It prints every event.

## 4 Instance appears

~1 min

**EC2 > Instances**, refresh:

- **Pending** then **Running**
- Lifecycle **spot**, zone ap-southeast-2...
- Spot Requests: **active / fulfilled**

## 5 Training starts

~1 min

- Dashboard state **running**
- Same instance ID as the console
- Epoch and step counting up
- A checkpoint about every 30 s
- S3 checkpoints/: refresh, files arrive

## 6 Let checkpoints build

~3 min

You can pause OBS here.

## 7 Real interruption

terminal 3

```
python scripts/demo_aws.py --interrupt
```

> **Say:** *"AWS has just told this machine it will be taken away in two minutes."*

Use the terminal, not the console: this account has no FIS console pages.

## 8 Notice and save

seconds

Point at each in the dashboard and terminal:

- **Interruption notice**: AWS's deadline (UTC, ends in Z)
- **Emergency decision**: time left, predicted times, size chosen
- **Emergency saved**: well under a second
- Worker exits with code **75**

Proof the FIS experiment ran (terminal 3):

```
aws fis list-experiments --region ap-southeast-2
```

Shows running, later completed.

## 9 Replacement

~80 s

- Console: old instance **Terminated**, a new one appears
- Dashboard: **Restored** at the same epoch and step
- New instance ID, **Restarts: 1**
- Training continues

## 10 Optional: again

+2-3 min

Repeat steps 7 to 9.

## 11 Finish

~25 min in

Dashboard: **completed**, accuracy about **80.7%**. The instance shuts itself down.

Pause OBS during the wait, resume a minute before the end.

## 12 Stop and check

OBS: **Stop Recording**. Then:

```
python scripts/demo_aws.py --no-follow
```

Must say no running instance.

## Editing

- Cut or speed up waiting. Caption it, e.g. **8x speed**.
- Keep the taskbar clock visible.
- **Never change a value on screen.**
- Final length: 5 to 8 minutes.

## If something goes wrong

> **No module named boto3**, or **spotsentry** not recognized

Not activated: `.venv\Scripts\Activate.ps1`

> **can't open file** scripts\...

Wrong folder: `cd D:\CCe1`

> **--start**: instance already running

Wait for that run to finish

> No instance after 3 min

EC2 > Auto Scaling groups > spotsentry-asg > Activity

> Dashboard can't read the store

Check `aws sts get-caller-identity`, restart the dashboard